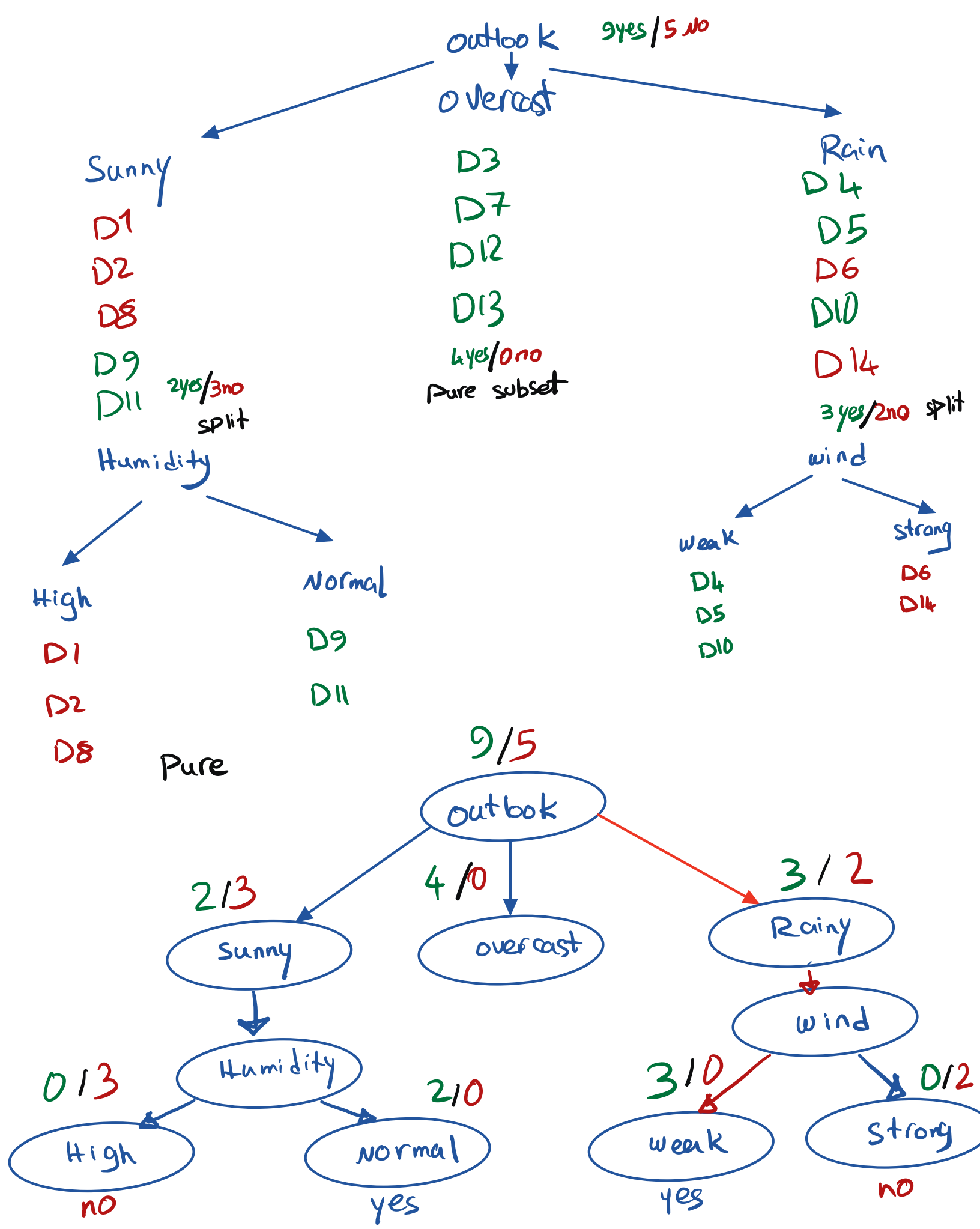
Predict if Parastoo will plays tennis (training examples: 9 yes / 5 NO

Day	Outlook	Humidity	wind	Play
D1	sunny	High	weak	NO
D2	Sunny	High	strong	NO
D3	overcast	High	weak	Yes
D4	Rain	High	weak	Yes
D5	Rain	normal	weak	Yes
D6	sunny	High	weak	NO
D7	overcast	normal	strong	yes
D8	Sunny	High	weak	NO
D9	Sunny	normal	weak	yes
D10	Rainy	normal	weak	yes
D11	sunny	normal	strong	yes
D12	overcast	High	strong	Yes
D13	overcast	normal	weak	yes
D14	Rainy	High	strong	NO
New data				
D15	Rainy	High	weak	?

Divide & conquer: - split into subsets - are they pure? (all yes or all no)
- if yes: stop if no: Repeat - see which subset new data falls into

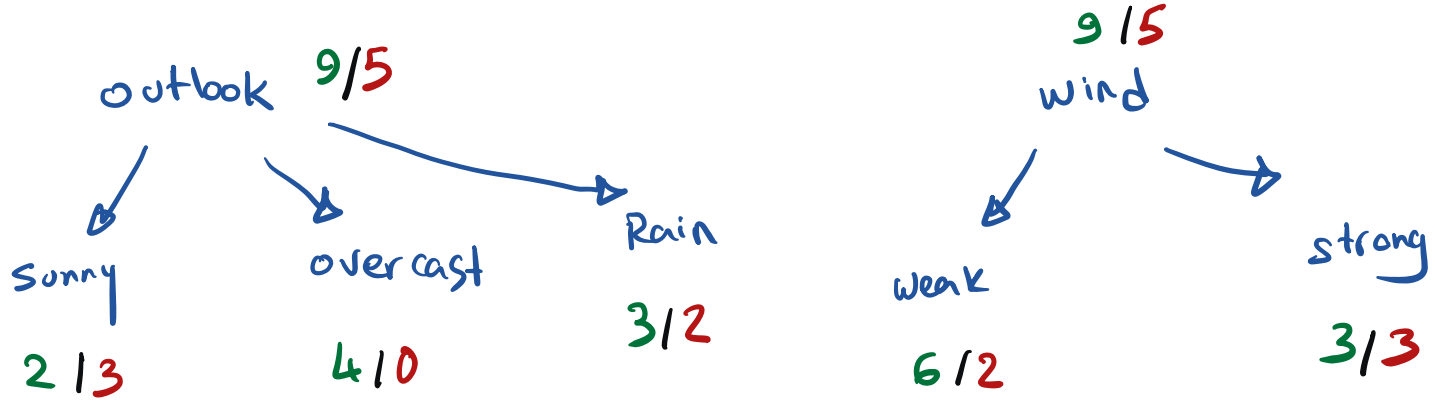

outlook
9yes/5no
overcast
Sunny
Rain
D3
D7
D12
D13
4yes/0no
Pure subset
D1
D2
D8
D9
D11
2yes/3no
split
Humidity
D4
D5
D6
D10
D14
3yes/2no
split
wind
High
Normal
D1
D2
D8
D9
D11
Pure
weak
strong
D4
D5
D10
D6
D14
9/5
outlook
2/3
4/0
3/2
Sunny
overcast
Rainy
Humidity
wind
0/3
2/0
3/0
0/2
High
normal
weak
Strong
no
yes
yes
no

Algorithm (1984 CART)

split (node, {examples}):

1. A ← the best feature for splitting the {examples}
2. Decision feature for this nod ← A
3. For each value of A, create new child node
4. split trainig to child nodes
5. For each child node/subset: if subset is pure: stop
   " " " not " : split

Which feature to split on?



Want measure "purity" of the split

Can't use P("yes"|set): must be symmetric: 4yes/0no as pure as 0yes/4no

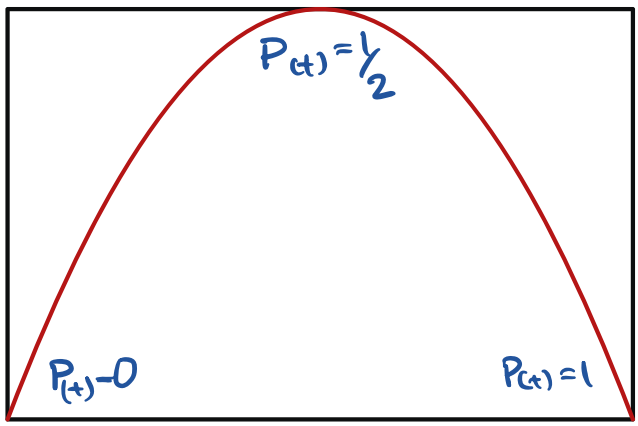
Entropy $H(S) = -P_{(+)} \log_2 P_{(+)} - P_{(-)} \log_2 P_{(-)}$

S: subset of training examples

$P_{(+)}/P_{(-)}$: % of positive / % of negative examples in S

Impure (3yes/3no) $H(S) = -\frac{3}{6}\log_2\frac{3}{6} - \frac{3}{6}\log_2\frac{3}{6} = 1$ bits

Pure set (4yes/0no) $H(S) = -\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4} = 0$ bits



Information Gain:

Want many items in Pure sets

Expected drop in entropy after split

$$Gain(S, A) = H(S) - \sum_{v \in values(A)} \frac{|S_v|}{|S|} H(S_v)$$

V: Possible values of A
S: set of examples $\{X\}$
S_v: subset where $X_A = V$

Mutual Information

- between feature A and class tables of S

wind 9/5

$H(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$

weak

6/2

$H(S_{weak}) = 0.81$

strong

3/3

$H(S_{strong}) = 1.0$

$$Gain(S, wind) = H(S) - \frac{8}{14}H(S_{weak}) - \frac{6}{14}H(S_{weak})$$

$$= 0.049$$

So we want maximise Gain

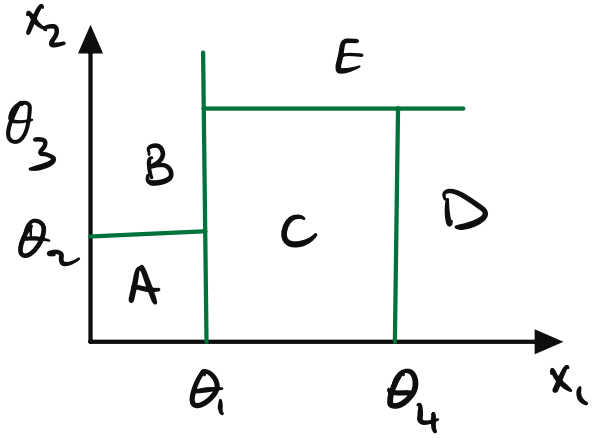
Issues:

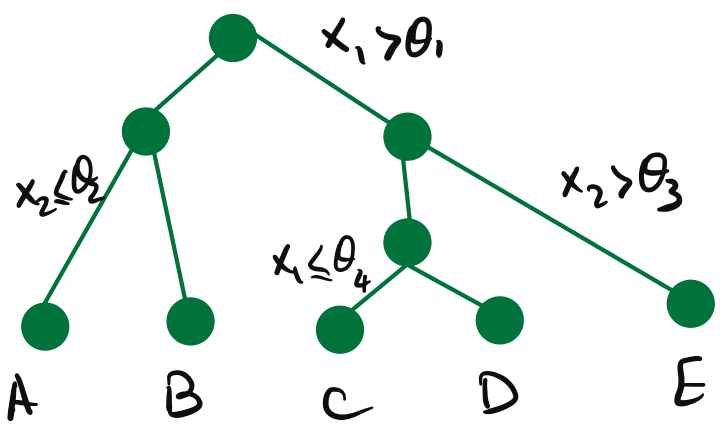
Overfitting ⟶ Pruning

What about numeric features?

create a split (Temperature > 72.3) = True, False

Threshold can be optimized





Multi class

- Predict most frequent class in the subset
- entropy $H(S) = -\sum_{(c)} P_{(c)} \log_2 P_{(c)}$, $P_{(c)}$ % of examples of class C in S

Regression

- Predicted output = average of the training examples in the subset
- ~~minimize Gain~~ minimize variance in subsets
- Can replace average by linear regression at the leaves